Resources Policy 85 (2023) 103778

Contents lists available at ScienceDirect

Resources Policy

journal homepage: www.elsevier.com/locate/resourpol

The impact of geopolitical relations on the evolution of cobalt trade network from the perspective of industrial chain

Yaoqi Guo [a,c], Yingli Li [b,**], Yongheng Liu [a], Hongwei Zhang [a,c,*]

[a] School of Mathematics and Statistics, Central South University, Changsha, 410083, China
[b] School of Business, Central South University, Changsha, 410083, China
[c] Institute of Metal Resources Strategy, Central South University, Changsha, 410083, China

ARTICLE INFO

Keywords:
Cobalt trade
Industrial chain
Geopolitical relations
Complex network
ERGM model

ABSTRACT

The turbulent international situation in recent years has aroused a profound impact on global cobalt trade, so it is necessary to investigate the influence of geopolitical relations on cobalt trade. This study divides the geopolitical network into cooperation network and conflict network. Besides, the evolution of global cobalt trade network is studied from the perspective of the whole industry chain. Furthermore, the impact of geopolitical relations on the evolution of cobalt trade network is studied based on QAP method and ERGM. The results show that (1) The scale of geopolitical cooperation network is larger than that of conflict network, while more and more countries have participated in the global geopolitical conflict in recent years; (2) The cooperation network has significant positive impact only on the evolution of midstream and downstream, and the conflict network has the opposite impact on them, indicating that geopolitical cooperation will promote the formation of new trade relations; (3) The influence on the evolution of cobalt trade network increases first and then decreases over time. This paper suggests that countries should pay special attention to geopolitical cooperation and establish more cooperation with countries with high political stability.

1. Introduction

Geopolitical relations and cobalt trade have consistently been closely intertwined. As recorded by history, the Angola civil war in 1975 and the Zaire riots in 1991 led to cobalt supply crisis (Catoto Capitango et al., 2022; Habib et al., 2016). During the Ukraine-Russia war, the traditional energy supply chain between Russia and Europe is interrupted, which led to the rise of energy prices. Therefore, investment in Europe is gradually inclined to clean energy. As a key metal material for energy transformation, cobalt will face more fierce competition (Nerlinger and Utz, 2022). In recent years, the geopolitical situation has changed dramatically, and its impact on the evolution of cobalt trade may become more complex due to the high risk of trade interruption. It is difficult to analyze the influence of geopolitical relations on cobalt trade only by relying on historical events. In addition, cobalt is becoming more and more significant due to the explosive growth of demand for electric vehicles. Therefore, in the context of the unpredictable international situation and the rapid development of the new energy industry, it is of great significance to analyze the impact of geopolitical relations on the evolution of cobalt trade.

With the frequent geopolitical conflicts in recent years and the gradual escalation of political conflicts, scholars began to consider the impact of geopolitical factors on the trade of cobalt and other critical metals. Li et al. (2022b) studied the impact of political risk on cobalt trade network pattern based on panel regression. Huang et al. (2021) explored the impact of political risk on tungsten trade competition. In addition, cobalt trade is a complex process involving multiple industrial chain stages and has different characteristics in each stage of the industrial chain. As the characteristics of different stages in the critical metal industry chain, geopolitics has different impacts on it. Li et al. (2022b) discovered that political risks have a positive impact on the midstream and downstream stage of cobalt trade but a negative impact on the upstream trade. Huang et al. (2021) found that political risks have the greatest impact on the midstream of tungsten trade but the smallest impact on the downstream trade. Although these studies revealed the fact that geopolitics can exert different influences on critical metals trade in different industrial chain stages, they only take politic risk into consideration and lack reflection on geopolitical

* Corresponding author. School of Mathematics and Statistics, Central South University, Changsha, 410083, China.
** Corresponding author.
E-mail addresses: guoyaoqi@csu.edu.cn (Y. Guo), yingli@csu.edu.cn (Y. Li), liuyongheng@csu.edu.cn (Y. Liu), hongwei@csu.edu.cn (H. Zhang).

https://doi.org/10.1016/j.resourpol.2023.103778
Received 28 January 2023; Received in revised form 19 May 2023; Accepted 1 June 2023
Available online 23 June 2023
0301-4207/

relations and structure. The single political risk index cannot systematically analyze geopolitical relations from the perspective of multilateral relations, which indicates that studying geopolitics from the perspective of relationship is necessary. Can geopolitical relations hinder or foster the evolution of cobalt trade network? Is there a heterogeneous effect on cobalt trade of geopolitical relations among different industrial stages? These are issues worthy of attention.

To study the influence of geopolitical relations on the evolution of cobalt trade network, the quantitative method of geopolitical relations must be clear. Geopolitics is often used to describe the means by which a country maintains its governance (Pollins, 1989), while power disputes and other events like regional independence, rebel groups, presidential election and some cooperation are considered as a part of geopolitics in recent years (Gong et al., 2022a; Gupta et al., 2019). Gong et al. (2022c) investigate the impact of four types of geopolitical risks (GPRs) on China's oil security. In pace with the advent of the big data era, quantitative methods for geopolitical relations are gradually increasing. Nejjari et al. (2021) investigate the relationship between cyber threats and geopolitics based on GDELT database. Li et al. (2022a) studied the spatio-temporal evolution and influencing factors of geopolitical relations in the near Arctic countries based on media news data. Levin et al. (2018) quantifying the intensity of geopolitical conflicts in all Arab countries after the Arab Spring by remote sensing and big data. Although the existing literature uses big news data to study geopolitical relations from a relationship perspective, it only studies from a single conflict perspective. Further, geopolitical relations are complex and not only contain negative conflict factors, but also have favorable cooperative relations (Bernauer and Böhmelt, 2020). The duality of cooperation and conflict is an important feature of geopolitics (Solingen, 2021). Therefore, it is more rigorous and comprehensive to study geopolitical relations from the perspective of cooperation and conflict.

This article has three main contributions: i) This article breakthroughs the perspective of measuring geopolitics mainly from single major event or risk index such as GPR index, and analyzes geopolitics from the relationship perspective, which is conducive to understand the global geopolitical situation comprehensively and identify the influential countries. Furthermore, this article divides geopolitical relations into cooperation and conflict and explores the heterogeneous effect on the evolution of cobalt trade network. ii) Considering that geopolitical relations may have different impacts on different stages of the cobalt industry chain, this study divides the cobalt industry chain into three stages to analyze the evolution of the trade pattern for cobalt more systematically, and explore the heterogeneous effect on cobalt trade of geopolitical relations among different industrial stages. This can help to make accurate suggestions on different industrial chain stages of cobalt. iii) In the context of energy transformation, cobalt resources are becoming increasingly important. This article mainly focusses on the impact of geopolitical relations on the evolution of cobalt trade. Most of the existing articles focus on the impact of geopolitical relations on energy trade and ignore cobalt. Given the rapid growth of cobalt demand in recent years and its importance to the new energy industry, it is necessary to incorporate the cobalt trade into geopolitical relations, which may help countries ensure the supply security and maximize benefits in the future when cobalt resources are getting scarcer.

Another part of this paper is arranged as follows. Section 2 is the theoretical mechanism framework. Section 3 describes the data and method used. Section 4 analyzes the evolution of geopolitical relationship networks and cobalt trade network. Finally, section 5 and section 6 give the empirical results of the ERGM model and conclusions respectively.

2. Theoretical mechanism framework

In this section, we analyze the influencing mechanism from two aspects (geopolitical cooperation and geopolitical conflict) to explore how the geopolitical relation influences the evolution of cobalt trade network at different industrial chain stages. The theoretical mechanism framework of the impact of geopolitical relations on cobalt trade is showed in Fig. 1.

(1) The influence and mechanism of geopolitical cooperation on cobalt trade network

Geopolitical cooperation includes signing trade cooperation agreements, international capital investment, etc (Sheng and do Nascimento, 2021). The geopolitical cooperation mainly affects the upstream of cobalt trade by signing trade cooperation agreements and international capital investment. The signing of trade agreements between countries is conducive to ensuring the stability of the supply of important resources, especially the supply of upstream raw materials like cobalt ore (Zuo et al., 2022). Regional trade agreements may generate technology spillover benefits (Martínez-Zarzoso and Chelala, 2021), this action can also help low-tech countries obtain high-tech trade products in the middle and lower industrial chain. Besides, international capital investment can exert both positive and negative effects on exporting country. For example, the RCEP (Regional Comprehensive Economic Partnership) can help stabilize the supply of chips and textiles upstream (Jiang and Yu, 2021). Foreign investment with additional conditions usually affects a country's autonomy in upstream raw materials allocation (Asiedu, 2006). Developed countries sometimes exchange investment for a country's main import right of mineral resources (Vivoda, 2011), and other countries can only compete for the remaining small amount of imports, which may cause more fierce competition for upstream raw materials.

(2) The influence and mechanism of geopolitical conflict on cobalt trade network

Geopolitical conflicts include war, political conflict, economic sanctions, technical blockade, etc (Monge et al., 2017). The war and political conflict will interrupt trade in all stages of the industrial chain (Ngoc et al., 2022). War is usually accompanied by traffic regulation, which leads to supply interruption in cobalt trade for all stages of the industrial chain (Jagtap et al., 2022). Geopolitical tensions may lead to trade restrictions or sanctions targeting specific countries or regions involved in cobalt production or trade, impacting the flow of cobalt across borders (Zheng et al., 2022). Technical blockade mainly affects the downstream of the industrial chain (Wei et al., 2020). The downstream of the industrial chain of many industries are technology intensive and it is easily affected by technical blockade. Although some developing factories have high-tech products, it does not mean they have key technologies. Technical blockade will increase the trade cost of the importer, and may even cause the import enterprise to be unable to carry out normal production (Meng and Zhao, 2022).

In summary, we try to reveal the theoretical influence mechanism of the association between the geopolitical relations and the evolution of cobalt trade network through two aspects, namely geopolitical cooperation and geopolitical conflict. In addition, we study different cobalt industrial chain stages at the same time and compare the differentiation effected by geopolitical relations. Our research might furnish new opinions and might uncover how geopolitical relations affects the evolution of cobalt trade network.

3. Data and method

3.1. Data

Cooperation and conflict are the dual characteristics of geopolitics. Events such as trade cooperation, joint military exercises and the signing of economic and trade agreements can all be defined as geopolitical cooperation. Events such as war, political conflict and technical blockade are defined as geopolitical conflicts (Monge et al., 2017; Sheng

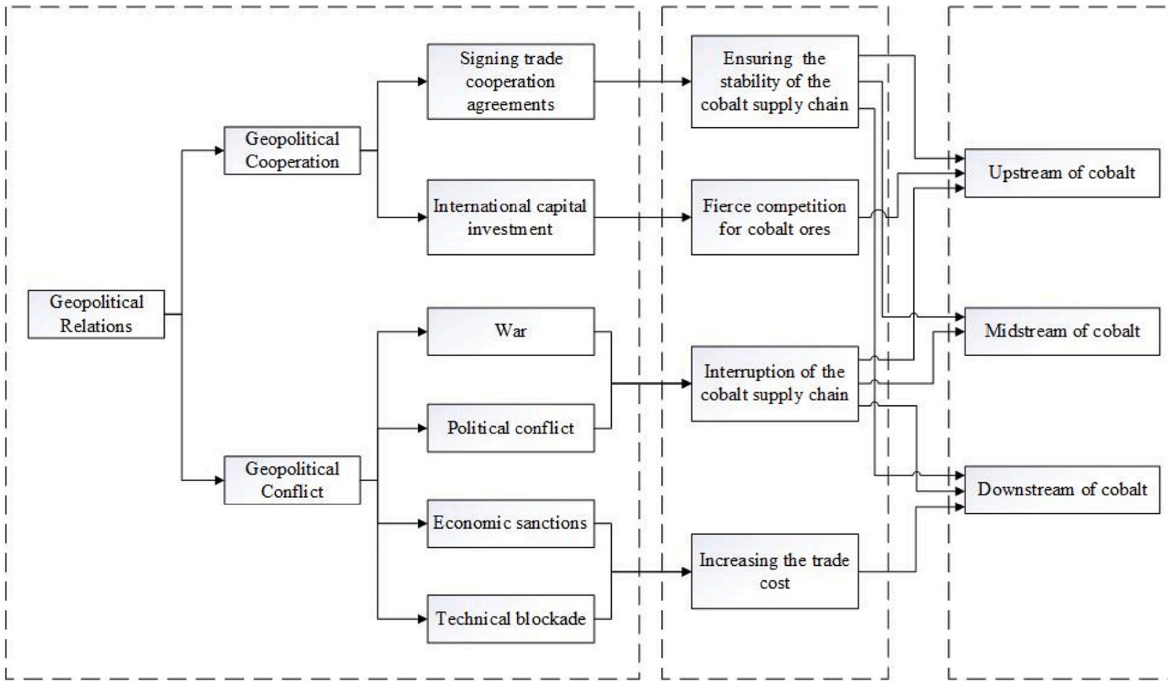

Fig. 1. Theoretical mechanism framework.

and do Nascimento, 2021). News big data can more comprehensively record geopolitical cooperation and conflict events between countries (Federo and Bustamante, 2022; Saz-Carranza et al., 2022). Therefore, followed by Lee et al. (2022) and Li et al. (2022a), this paper quantifies geopolitical relations based on news information data. GDELT is a news information database supported by Google jigsaw, which can track and record events released in various forms such as global TV, website and radio in real time (Li et al., 2022a). This article selects the event database of GDELT from 2011 to 2020 because the database keeps events that all reflect geopolitical cooperation or conflict. Events recorded in this database can be classified into four categories, namely, the verbal cooperation and practical cooperation, verbal conflict and practical conflict. This article calculate the total number of actual and verbal cooperation or conflict events as the cooperation intensity, and countries participating in cooperation or conflict are regarded as nodes, which is the basis of geopolitical relationships network.

The cobalt trade data from 2011 to 2020 is downloaded from United Nations Commodity Trade Database. With reference to Li et al. (2022b),

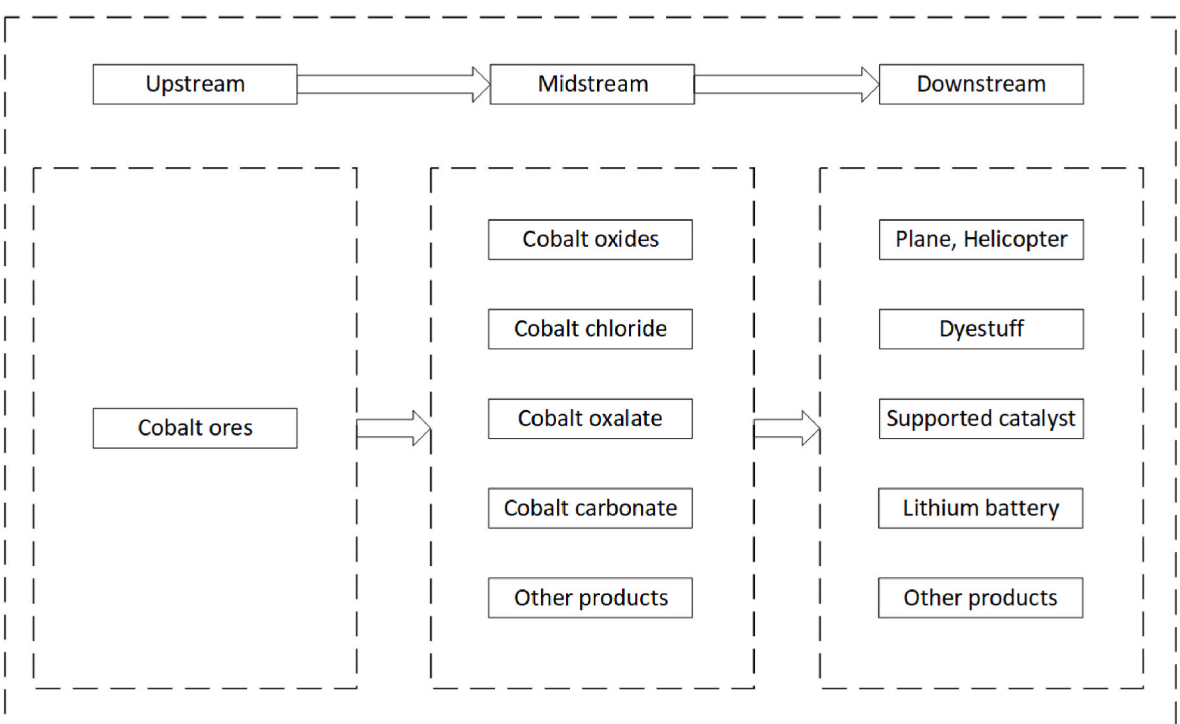

Fig. 2. Part of products in cobalt industrial chain.

we divide the cobalt industry chain into upstream, midstream, and downstream. According to Sun et al. (2019), the upstream is mainly cobalt ores, therefore we select cobalt ores and concentrations (HS 260500) to represents upstream product of cobalt. Considering the availability of data and the situation in midstream of cobalt, we choose 8 product such as cobalt chloride and cobalt oxalate to represents midstream product of cobalt. Following Shi et al. (2022), the downstream products of cobalt mainly include helicopters, airplanes, lithium batteries, etc. Thus, we choose 76 types of products that closely related to cobalt consumption to represent downstream product of cobalt. Table A1(in the appendix) shows the list of products selected in this research. The ISO country code is used to denote each country, and Table A2 (in the appendix) shows the corresponding relationship between ISO code and country name. Fig. 2 shows the part of products in cobalt industrial chain. The trade data of the midstream and downstream stage are the sum of the trade values of these all product we selected with reference to Huang et al. (2021).

In addition, the country attribute data used in this paper, such as GDP, population, and foreign direct investment are from the World Bank database. The relations data like geographic distance, common geographical boundary, and common official language are from the CEPII database.

3.2. Method for constructing the trade network and geopolitical relationships network

Using complex network theory, this article constructs weighted trade network $G_1 = (V_1, E_1)$ and geopolitical relationships network $G_2 = (V_2, E_2)$. In the trade network, the cobalt importer $V_1 = \{v_1, v_2, v_3, \ldots, v_n\}$ represents the trade network nodes; The import relationship $E_1 = \{e_1, e_2, e_3, \ldots, e_n\}$ represents the trade network connection, and the trade volume of an import transaction is the weight of the edge. We divide the geopolitical network into cooperation and conflict networks. Countries in the world $V_2 = \{v_1', v_2', v_3', \ldots, v_n'\}$ denotes the geopolitical relationships network nodes; The cooperation and conflict between countries $E_2 = \{e_1', e_2', e_3', \ldots, e_n'\}$ denotes the geopolitical relationships network connection, and the intensity of cooperation and conflict are the weight of the edge.

The topological structure of a network is defined as follows.

(1) Degree and weighted degree

The degree of a node i refers to the total number of neighbor nodes directly connected to the node i in the network, and the average degree $\overline{k}$ is the average value of all the nodes' degrees k_i. w_i is the weighted degree.

$$k_i = \sum\nolimits_{j=1}^{n} e_{ij} \quad i = 1, \ldots, n \tag{1}$$

$$\overline{k} = \frac{\sum k_i}{N} \quad i = 1, \ldots, n \tag{2}$$

$$w_i = \sum\nolimits_{j=1}^{n} e_{ij} w_{ij} \tag{3}$$

$$\overline{w} = \frac{\sum w_i}{N} \tag{4}$$

where N represents the number of nodes in the trade or geopolitical relationships network each year, w_{ij} is the trade volume or intensity of cooperation and conflict between nodes i and j, and $\overline{w}$ is the average weighted degree of the network.

(2) Average path length

Average path length is an index to reflect closeness of network, it is applied to calculates the average length between nodes in the network.

$$L = \frac{1}{N(N-1)} \sum_{ij} d(i, j) \tag{5}$$

where $d(\alpha_i, \alpha_j)$ is the shortest distance between nodes i and j.

(3) Density

Density can reveal the nearness of the importer's relationships in network. It is the proportion of edges with the same number of nodes in the network to the number of all edges.

$$D = \frac{2M}{N(N-1)} \tag{6}$$

where M denotes the network's actual competitive relationships, N is the number of nodes.

(4) Average clustering coefficient

The tightness of competition among countries can be expressed by cluster coefficient, and its value ranges from 0 to 1. The degree of network tightness is positively correlated with the size of clustering coefficient. Average clustering coefficient has same function as cluster coefficient, the definition is as follows(Barrat et al., 2004):

$$C_i = \frac{2n_i}{k_i(k_i - 1)} \tag{7}$$

$$\overline{C} = \frac{1}{N} \sum\nolimits_{i=1}^{N} C_i \tag{8}$$

where k_i denotes the degree of node i, and n_i is the actual number of connections among the neighbors of node i.

(6) Eigenvector centrality

Eigenvector centrality used to measure the connection between nodes and important countries. If one economy establishes a link to countries with a central status in network, it will get higher value of eigenvector centrality. We calculate this indicator as follows:

$$EC_i = \frac{1}{\lambda} \sum_{j \in N_i} w_{ij} EC_j \tag{9}$$

where λ is a constant, N_i is the set of nodes that link to node i, and w_{ij} indicates the total trade volume from i to j.

3.3. QAP correlation coefficient

The QAP (Quadratic Assignment Procedure) correlation coefficient is a measure of similarity or association between two matrices, typically used in network analysis, social network analysis, and other related fields. The QAP correlation coefficient is calculated using the following formula:

$$coeff = \frac{s - E[s]}{sqrt(Var[s])} \tag{10}$$

where s is the observed similarity or association score between the two matrices being compared. $E[s]$ denotes the expected similarity or association score under the null hypothesis, which is calculated as the average of similarity scores obtained from randomly permuting the rows and columns of one of the matrices while keeping the other matrix fixed. This is typically done through multiple random permutations (e.g., 1000 or more) to obtain a distribution of expected similarity scores. $Var[s]$ is the variance of the similarity scores obtained from the random permutations.

A higher QAP correlation coefficient indicates a stronger similarity or association between the two matrices, while a lower coefficient indicates a weaker similarity or association. The coefficient ranges from −1 to 1, with 1 indicating perfect similarity or association, −1 indicating perfect dissimilarity or association, and 0 indicating no similarity or association.

3.4. ERGM conceptualization

The ERGM can test the existence of links in networks, which based on the assumption that links are dependent on each other.

The weighted cobalt trade network matrices can be denoted as $y = [y_{ij}]$, and the Pr $(Y = y|\theta)$ represent the probability of y appearing in the possible set Y under the condition θ. According to Xu et al. (2021), assuming that the probability of observed international trade network (y) depends on different kinds of network structure statistics, the general form of the ERGM model can be expressed as follows:

$$Pr(Y=y|\theta)=\frac{1}{\kappa}exp\left\{\theta_{\alpha}^{T}z_{\alpha}(y)+\theta_{\beta}^{T}z_{\beta}(y,X)+\theta_{\gamma}^{T}z_{\gamma}(y,Z)\right\} \tag{11}$$

Where κ denotes constant, which is mainly used to ensure that the model has an appropriate probability distribution. $z_{\alpha}(y)$ is a series of endogenous network structure variables that may affect the evolution of the cobalt trade network. $z_{\beta}(y,X)$ is a series of exogenous network variables that affect the structure of the cobalt trade network. $z_{\gamma}(y,Z)$ denotes a series of node attribute variables. θ_{α}^{T}, θ_{β}^{T} and θ_{γ}^{T} denote the estimation parameter vectors of endogenous network structure variables, exogenous network variables, and node attribute variables, respectively. If these estimated parameters can meet the test of statistical significance, it shows that the variables have an important influence on the evolution of cobalt trade network.

3.5. Variables in the ERGM

ERGM model generally assumes that the network relationship is mainly affected by three types of variables, including endogenous network structure variables, exogenous network variables effect and exogenous node attributes variables. The common main statistics are shown in Table 1.

(1) Endogenous Network Structure Variables in the ERGM.

The pattern of self-organization of network relations is called “pure structure effect”, also known as “endogenous network structure variables”. In this article, an endogenous network structure variables named edges are selected to enter the model.

(2) Exogenous Network Variables in the ERGM

The evolution of the cobalt trade network is affected by the external dual relationship besides the pure structure effect. ERGM can include network variables in the analysis to verify the interdependence of different types of binary relationships. This article mainly care about two kinds of geopolitical networks, namely, the cooperation network (Edgecov(COO)) and conflict network (Edgecov(CON)). In addition, there are other factors that can affect the evolution of cobalt trade networks. Spatial distance can affect the trade efficiency, and the closer the distance is, the easier it is to form trade relations (Allen and Arkolakis, 2014). There are relatively few communication barriers between countries that use the same official language, and the reduction of communication barriers is conducive to the establishment of trade relations (Fink et al., 2005). The land border between the two countries means that they are closer to each other, and they can use lower cost transportation. At the same time, it has been proved that most countries bordering on land in the world have trade relations (Hoekman and Nicita, 2011). Therefore we add geographic distance network (Edgecov (DIST)), common official language network (Edgecov(COL)), and common geographical boundary network (Edgecov (CGB)) in ERGM model. If two countries are connected by land, the edge of the CGB network is assigned a value of 1. If the two countries are not connected by land, the edge of the network is assigned a value of 0. Likewise, if two countries have the same official language, the edge of the CGL network is assigned a value of 1. If the two countries do not share the same official language, the edge of the network is assigned a value of 0. We use adjacency weighted matrixes to denote the different humanistic and geopolitical relationships.

Table 1
The variables used in ERGM.[a]

Category	Variable	Variable meaning
Endogenous network structural variables	Edges	Number of links in the cobalt trade network
Exogenous network variables	Edgecov (COO)	Geopolitical cooperation between countries
	Edgecov (CON)	Geopolitical conflict between countries
	Edgecov (DIST)	Geographical distance between countries
	Edgecov (COL)	If the two countries use the same official language, the edge of the network is 1, otherwise, it is 0
	Edgecov (CGB)	If the two countries connected by land, the edge of the network is 1, otherwise, it is 0
Node attribute variables	Nodecov (GDP)	Gross domestic product
	Nodecov (FDI)	Foreign direct investment
	Nodecov (POP)	Resident population
	Nodecov (dcoo)	Degree of countries in cooperation network
	Nodecov (dcon)	Degree of countries in conflict network
	Nodecov (wdcoo)	Weighted degree of countries in cooperation network
	Nodecov (wdcon)	Weighted degree of countries in conflict network

[a] Data source: The GDP, FDI and POP come from World Bank database. The DIST, COL and CGB come from CEPII database. Other variables are calculated from network that we construct in this article.

(3) Exogenous Node Attributes Variables in the ERGM.

A country’s resource endowment and advantages will affect its behavior and network evolution. The node attribute effect mainly includes homogeneity and Matthew effect. Homogeneity is that countries with the same characteristics are likely to establish relations, and Matthew effect refers to the phenomenon that the strong stronger and the weak weaker. This paper focuses on the influence of population (Nodecov(POP)), GDP (Nodecov(GDP)), and foreign direct investment (Nodecov(FDI)). For the Matthew effect of the centrality of geopolitical relationships network, the degree and weighted of cooperation network (Nodecov(COO)) and conflict network (Nodecov(CON) are incorporated in ERGM model.

4. Overall analysis of networks

4.1. Overall analysis of the geopolitical relationship network

In this paper, the geopolitical relationship network is divided into cooperation and conflict networks. Considering the epidemic situation and data range, we shows the diagram of geopolitical cooperation network and conflict network for 2019 in Fig. 3. We adopt the ISO code to represent the country in network, and the font size represents the

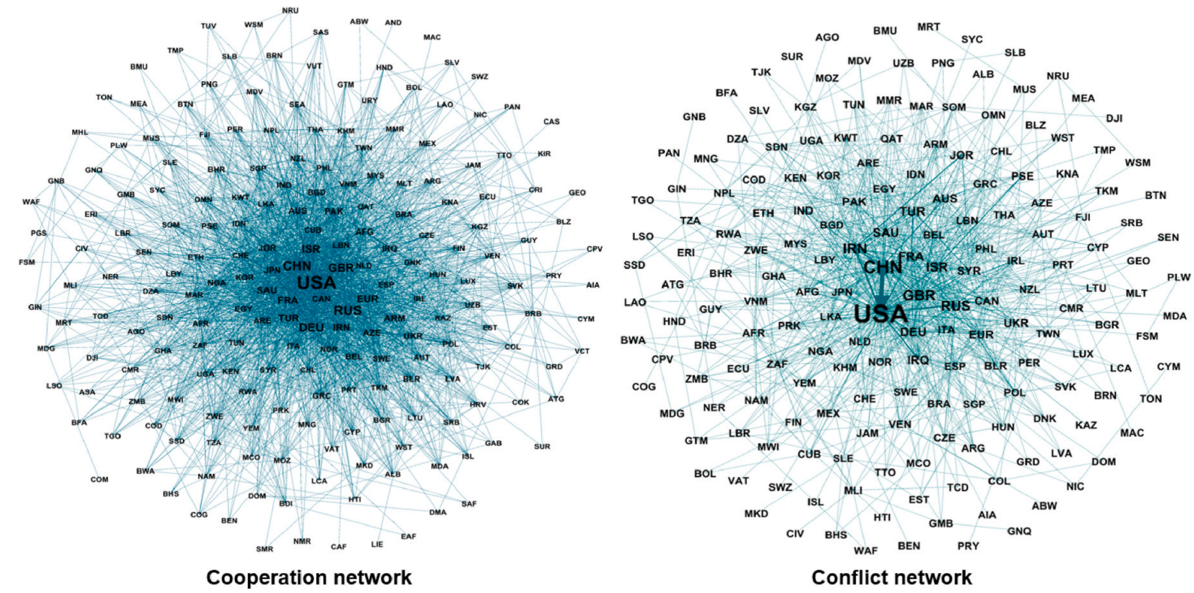

Fig. 3. The diagram of geopolitical relations network in 2019.
USA are most important country in both cooperation network and conflict network.

weighted degree of the country. The thickness of edge denotes the number of cooperation events or conflict events between two countries, and we call them as cooperation intensity or conflict intensity. The United States remains the most important country in cooperation network and conflict network at the same time, followed by China, the United Kingdom, and German. In addition, the cooperation network is more intensive, indicating that geopolitical cooperation events are more frequent than conflict events.

To study the number of countries that participate in geopolitical cooperation and conflict, Fig. 4 shows the number of nodes and edges in two networks. The number of nodes and edges of the cooperation network is higher than that of the conflict network every year, which indicates that more countries have participated in cooperation events than conflict events during this period. It can be seen that cooperation is the strategic theme of the development of all countries. In addition, the number of nodes in conflict network rose after 2013, denoting that more and more countries have joined in geopolitical conflicts.

The other indicators of these two geopolitical relationship networks are calculated in Table 2. The network density indicates the interaction of countries participating in cooperation events or conflict events, and the maximum value of 1 indicates that all countries have interaction. The density of cooperation network is higher than that of conflict network, which indicates that the interaction of countries participating in cooperation events is greater than that of participating in conflict events. In addition, the average shortest path of the network is gradually decreasing, the clustering coefficient is gradually increasing, and the average degree and reciprocity are gradually increasing, which indicates that international cooperation is closer.

To identify the most important countries in the cooperation and conflict networks, Fig. 5 shows the top 10 of degree, weighted degree, and eigenvector centrality. The United States remained the largest degree and weighted degree in cooperation network and conflict network, which indicates that the United States has established the most cooperative relationships and also has conflict relations with many countries. Further, the United States kept the largest eigenvector centrality in both two networks during this period, followed by the United Kingdom, denoting that these two countries had a high and stable position in the geopolitical relationship network from 2011 to 2020. Moreover, some countries are prominent in certain years. For example, the Syria had a high level of weighted degree for conflict network in 2011 which may be caused by the war between Syria and Libya in 2011, and Israel had a high level of weighted degree in conflict network which may be caused by a series of political events that triggered by the Israeli general election in 2013.

4.2. Overall analysis of the cobalt trade network

Fig. 6 shows the cobalt trade network diagrams of different industrial

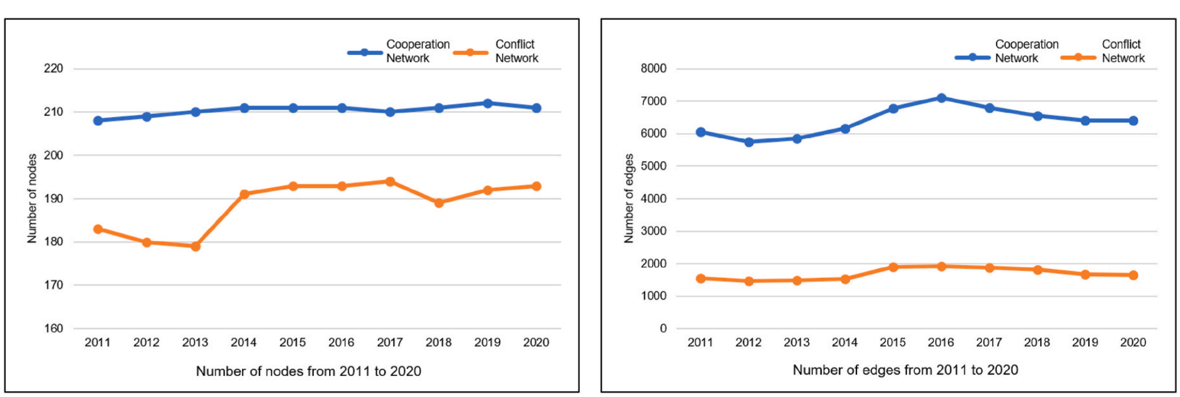

Fig. 4. The number of nodes and edges in geopolitical networks.

Table 2
Other indicators of geopolitical relationship networks.[a]

	Cooperation network					Conflict network				
Indicator/Year	2011	2013	2015	2017	2019	2011	2013	2015	2017	2019
Density	0.14	0.13	0.14	0.15	0.14	0.05	0.05	0.04	0.05	0.05
Average shortest path	2.01	1.98	1.96	1.95	1.97	2.47	2.46	2.39	2.35	2.42
Clustering coefficient	0.44	0.41	0.42	0.45	0.43	0.29	0.27	0.26	0.30	0.29
Average degree	58.22	55.65	58.47	64.15	60.41	17	16.57	15.92	19.72	17.48
Reciprocity	0.85	0.86	0.87	0.87	0.85	0.53	0.55	0.55	0.56	0.54

[a] These indicators mainly reflect the tightness of the network.

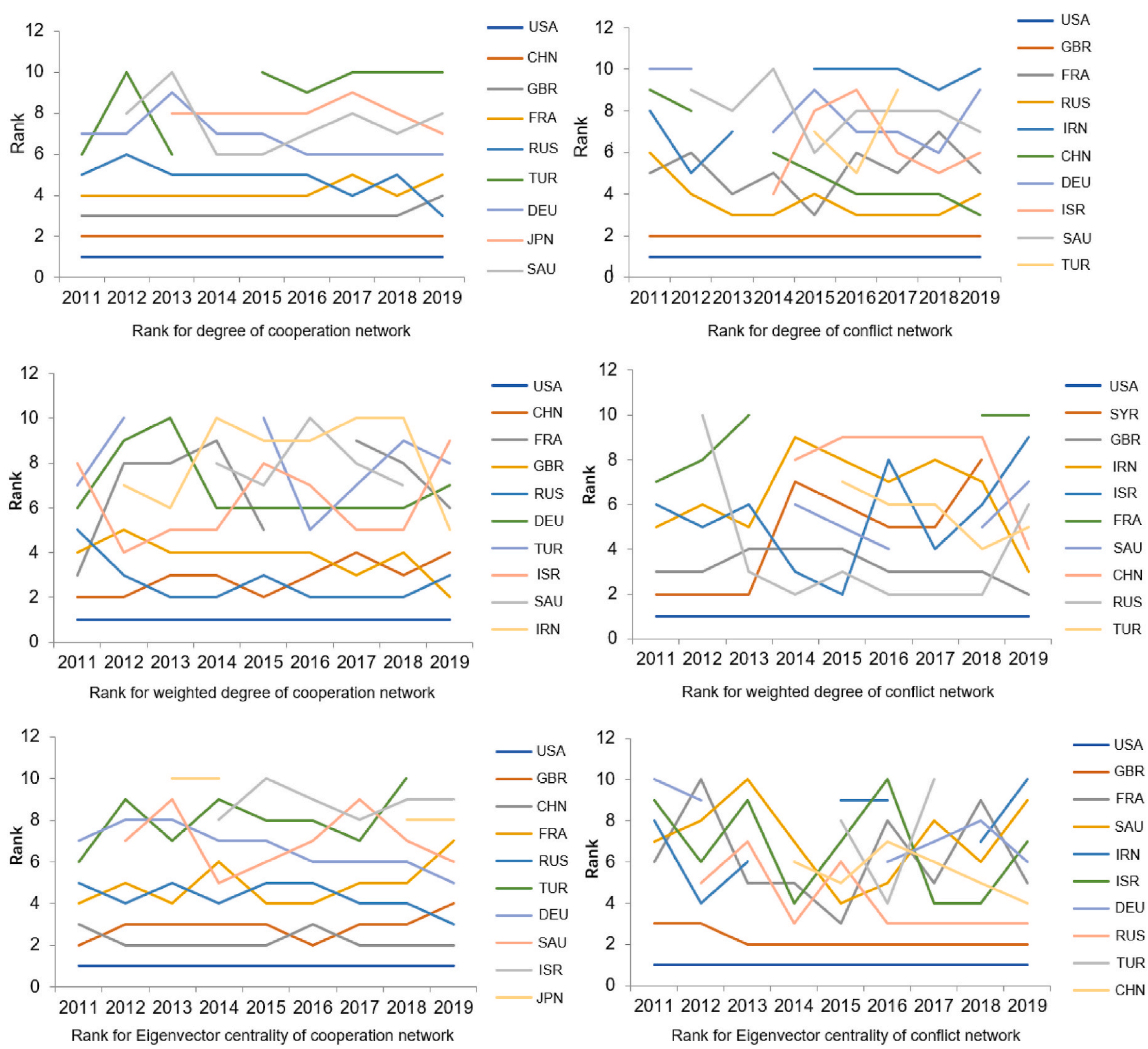

Fig. 5. The top 10 of three degree of centrality in geopolitical relations networks.
These figures mainly reflect the important countries and their importance rankings in the geopolitical relationship network in different years from three indicators.

stages for 2020. The letters in the figure represent the ISO code of the country, and the font size represents the weighted degree of the country. The larger the font of the country, the greater the weighted degree. The thicker an edge is, the larger the trade volume of cobalt products between importing countries. Obviously, the trade pattern of upstream is the simplest, while that of downstream is the most complicated. In 2020, Congo plays a very important role in cobalt upstream and midstream trade, and China, the United States and Germany are important members in the downstream.

Fig. 7 shows the evolution of the cobalt trade network scale from 2011 to 2020. The number of nodes in the cobalt trade network remained relatively stable after 2011, while the number of edges in upstream has dropped after 2018 and this decline is especially obvious in 2020. This situation may be caused by the sharp rise in cobalt prices has compressed the profits of downstream products, which forces some downstream trading countries to withdraw from the market. In addition, the epidemic of COVID-19 might be an important reason for the sharp decline in 2020.

To further understand the dynamic evolution of network characteristics, Fig. 8 shows the average degree and average weighted degree from 2011 to 2020. The average weighted of the upstream trade network increased from 2014 to 2018, which was benefited from the rise of cobalt price and the sharp increase of cobalt demand. The price fluctuation of cobalt is an important influencing factor in cobalt trade (Gong et al.,

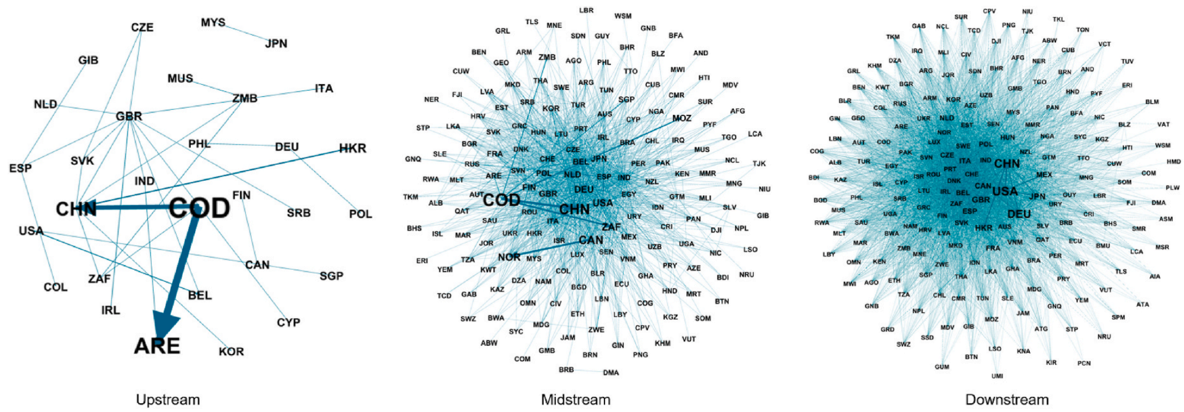

Fig. 6. The network diagrams of cobalt trade in 2020.

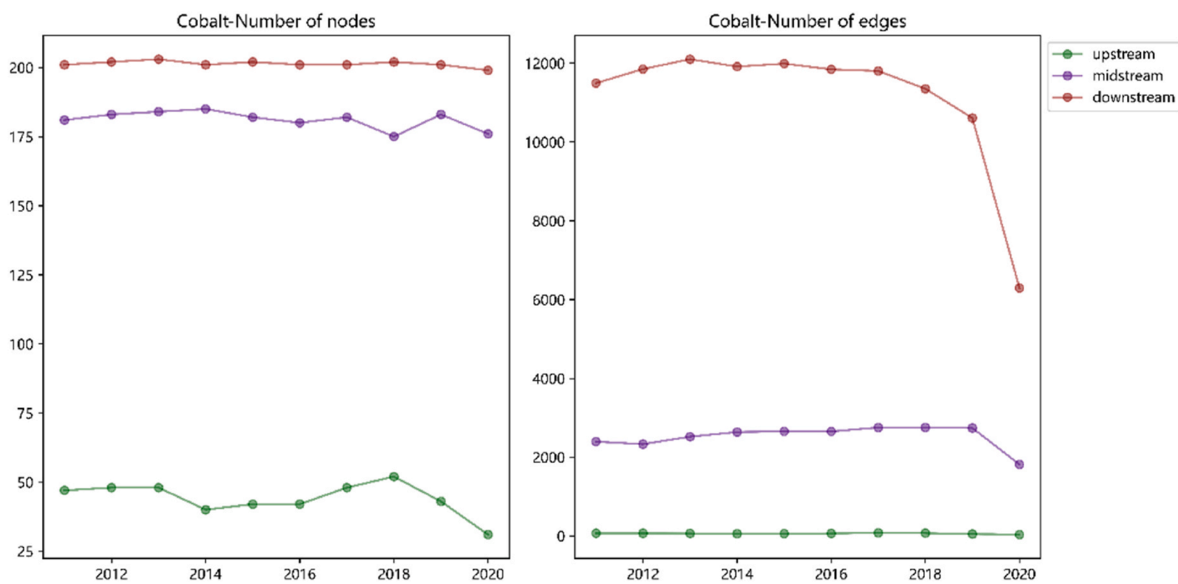

Fig. 7. The evolution of the cobalt trade network scale.

2022b; Gong and Lin, 2021). In addition, the average weighted of the midstream and downstream trade network dropped sharply after 2019 while that of upstream kept relatively stable. That is because the midstream and downstream markets are extensive, and the impact of the epidemic of COVID-19 is greater (Wen et al., 2021).

The average path length can reflect the trade transmission efficiency of the trade network, and Fig. 9 shows the average path length of cobalt trade network between 2011 and 2020. The average path length of upstream trade network fluctuates greatly, ranging from 1.6 to 4.1. Namely, each upstream trade chain cross 1.6–4.1 intermediary countries on average. Each midstream trade chain passes through 2.3 intermediate countries on average, and that of downstream passes through 1.6 intermediate countries on average, which indicates that the downstream trade is more efficient and stable. What's more, the average path length of the cobalt upstream trade network decreased significantly after 2016. That is may because the global supply of raw materials for cobalt upstream has increased significantly and improve the transmission efficiency.

The network density and average clustering coefficient can reflect the closeness of trade network, and Fig. 10 shows the evolution of these two indicators. The network tightness of cobalt upstream and midstream trade network is not strong, with the network density below 0.1. This means that there is still much room for cooperation among countries. The density of the cobalt upstream trade network mainly fluctuates around 0.28, and drops sharply after 2018, indicating that the global cobalt upstream trade market has undergone drastic changes after 2018. Further, the evolution trend of average clustering coefficient is similar to that of network density, and it dropped after 2018 too.

There exist regional areas through long-term evolution which is called community. Fig. 11 shows the division of global cobalt trade network community, and countries with the same color represent the same community. There are mainly two major communities in the global cobalt upstream trade network, and the core members are Congo, China, and the United Kingdom. Cobalt midstream trade network is mainly divided into four groups. The largest group is mainly composed of Germany, the United States, the United Kingdom, the Netherlands and other countries, and the second is composed of important exporting countries such as Congo and China. There are many communities in cobalt downstream trade network, mainly including three large communities. The largest community is mainly composed of the United States and China, and the second largest community is mainly composed of European countries with Germany as the core. It indicates that Congo and China are the most important countries in upstream and midstream of cobalt trade, and the United States and some European countries keep significant role in that of downstream.

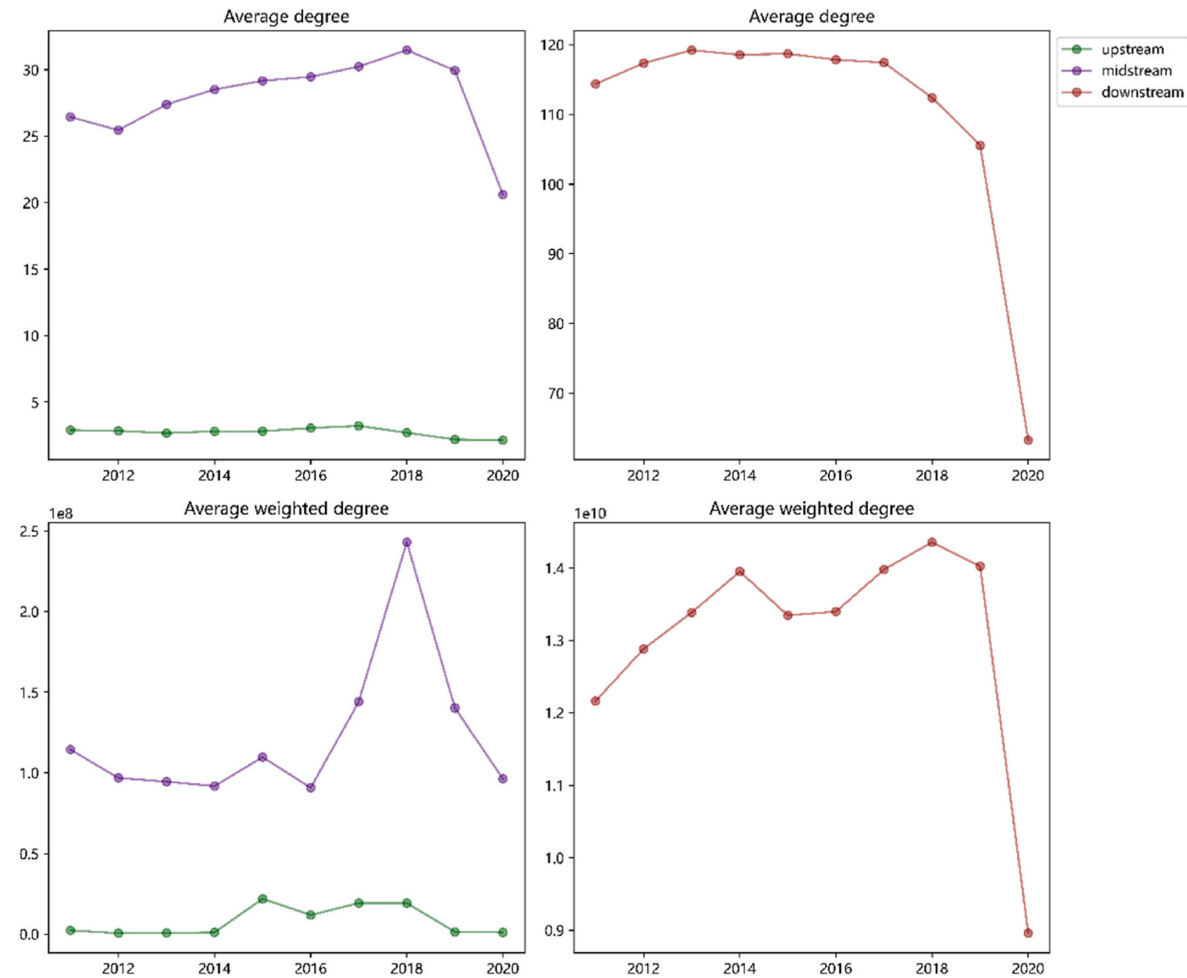

Fig. 8. The average degree and average weighted degree of cobalt trade network.

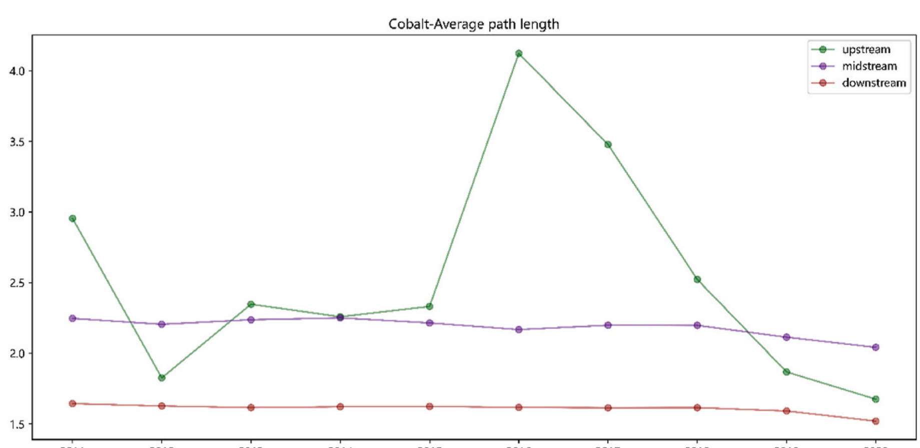

Fig. 9. The average path length of cobalt trade network.

5. Analysis of the influence of geopolitical relations on the evolution of cobalt trade network

5.1. QAP correlation analysis

This article applied the QAP method to analyze the correlation between geopolitical relationships network and cobalt trade network. QAP is a non-parametric test method based on random permutations to test the correlation between two matrices. As all rows and columns of the matrix are identically permuted in the QAP method, the problem of auto-correlation is addressed. Based on the constructed geopolitical relationship network and cobalt trade network, QAP correlation coefficient is shown in Table 3.

In general, the correlation between the global cobalt trade network and the cooperation network is stronger than that between the global cobalt trade network and the conflict network. From the perspective of

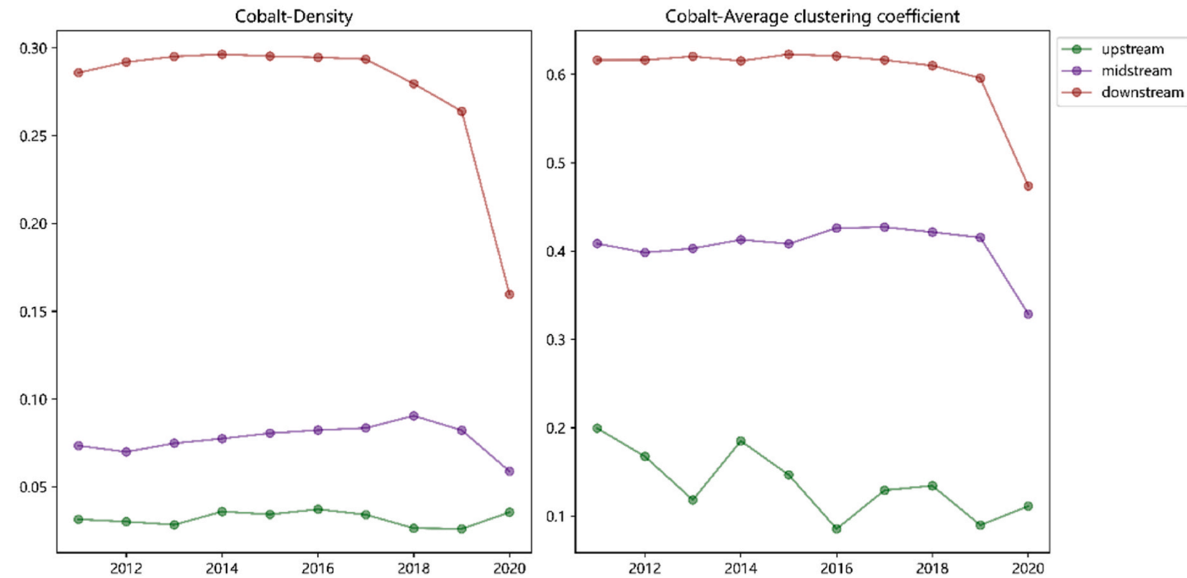

Fig. 10. The network density and average clustering coefficient of cobalt trade network.

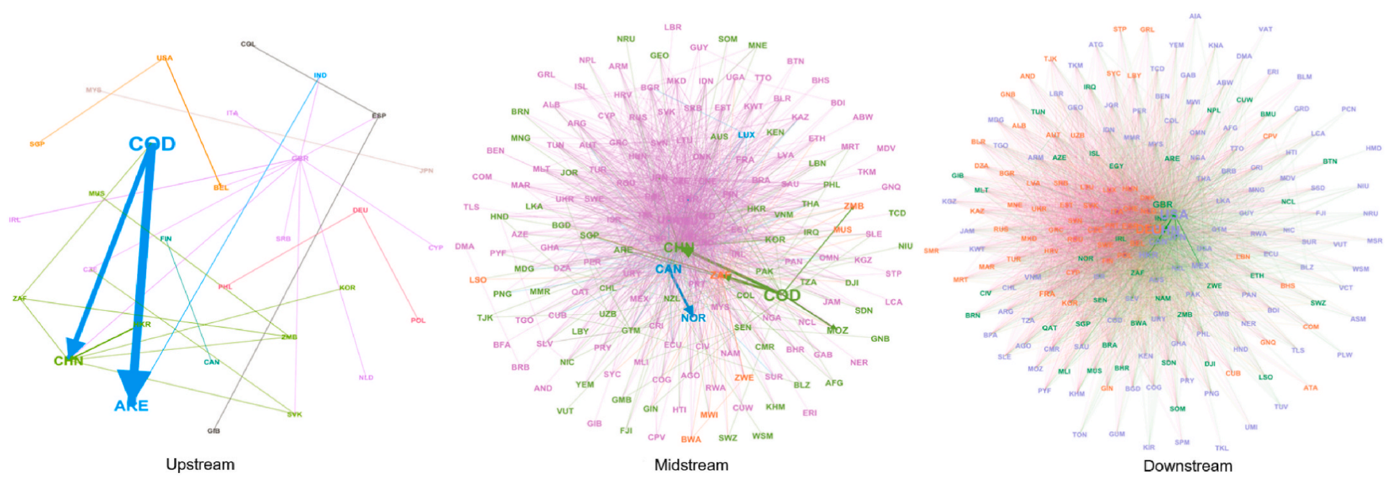

Fig. 11. The division of global cobalt trade network community.

Table 3
The QAP correlation coefficient between geopolitical relations network and cobalt trade network.

Cobalt-Upstream										
Year	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Cooperation	0.062	0.031	0.234***	0.192***	0.040	0.076	0.409***	0.143***	0.089*	0.391***
Conflict	0.061	0.074	0.238***	0.236***	0.059	0.052	0.368***	0.104***	0.087*	0.229***
Cobalt-Midstream										
Year	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Cooperation	0.089***	0.092***	0.093***	0.126***	0.101***	0.183***	0.129***	0.226***	0.258***	0.394***
Conflict	0.038***	0.069***	0.071***	0.079***	0.055***	0.096***	0.078***	0.092***	0.171***	0.353***
Cobalt-Downstream										
Year	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Cooperation	0.496***	0.372***	0.360***	0.362***	0.400***	0.456***	0.378***	0.379***	0.355***	0.448***
Conflict	0.181***	0.210***	0.266***	0.166***	0.235***	0.197***	0.183***	0.230***	0.329***	0.466***

Notes: *, **, and *** indicate significance levels of 10%, 5%, 1%, respectively. The value is correlation coefficient.

industrial chain, the correlation coefficient between cobalt upstream trade network and geopolitical relationship network is not significant in many years. The correlation coefficient between cobalt downstream trade network and geopolitical relationships network is the largest, followed by cobalt midstream. It means that the geopolitical network has the closest relationship with cobalt downstream trade network, and has the least close relationship with the upstream network.

5.2. Analysis of the ERGM model results

Based on QAP correlation analysis, we can further analyze the impact of geopolitical relationship on the evolution of cobalt trade network through the estimation of the ERGM. Considering the impact of the COVID-19, there may be drastic fluctuations in trade data, and we choose 2019 for ERGM estimation. Table 4 represents the estimation results of the ERGM model. In basic model, we only considered the endogenous network structure effect and the exogenous node attributes effect, while in proposed model, we added the exogenous network variables. Comparing the results of basic model and proposed model, we find that the AIC and BIC values of proposed model are smaller than basic model, which means that proposed model is better than basic model.

From the estimation result of proposed model, the coefficients of cooperation network (Edgecov(COO)) are positively significant in midstream and downstream, and that of upstream is not significant. This shows that cooperation between countries can promote the establishment of cobalt midstream and downstream trade, but it has no significant impact on the upstream trade of cobalt. The coefficients of conflict network (Edgecov(CON)) are negatively significant in midstream and downstream, and that of upstream is not significant. It means that conflict between countries will impede the cobalt midstream and downstream trade, but the effect on upstream is not significant.

The above results indicate that different geopolitical relationships have different effects on the evolution of cobalt trade network. Possible explanations are as follows. Cooperation between countries usually promotes bilateral trade, such as the Belt and Road initiative (BRI). Geopolitical cooperation like BRI can significantly reduce transportation time and trade costs, and promote financial cooperation (de Soyres et al., 2019), which has laid a foundation for the establishment of cobalt trade relations between the two countries. Geopolitical conflict like Russia - Ukraine War may the hinder the establishment of trade relations (Alam et al., 2022). These conflicts can disrupt the normal trade contract, force capital flight, and affect investors' confidence (Abbassi et al., 2022), which is not conducive to the formation of cobalt trade relations.

The result also reveal that the influence of geopolitical relationship is different in different industrial chain stages of cobalt trade. In terms of the second heterogeneity, no matter the cooperation network or the conflict network, they only have a significant impact on the evolution of trade relationships in the midstream and downstream of cobalt products, but have no significant impact on that of upstream. According to Gustafsson et al. (2022), the upstream market of cobalt is highly concentrated, which means that there are fewer countries participating in cobalt upstream trade. The cooperation or conflict relations among these major upstream trade participants account for a small proportion in the complex geopolitical relations among countries in the world. Therefore, it may be a possible explanation that geopolitical cooperation and conflict networks have no significant impact on the evolution of cobalt upstream trade. Further, most cobalt upstream producing countries are located in countries with medium or extremely weak political stability, and conflict events are relatively frequent (Habib et al., 2016). Therefore, these countries are relatively insensitive to small-scale conflicts between countries (Lohner et al., 2019), and the impact of geopolitical conflicts on the trade of cobalt upstream is correspondingly weakened. Compared with the upstream market of cobalt, the middle and downstream markets are widely distributed. The production countries are all over the world, and the number of exporting countries is far more than that of upstream market (Li et al., 2022b). Moreover, the demand side of the middle and downstream products is also more extensive. Therefore, the midstream and downstream of cobalt trade are more vulnerable to geopolitical conflicts and cooperation.

To further test the dynamic impact effect of the network, Fig. 12 shows the evolution of the impact coefficient of the geopolitical relationship network (including cooperation network and conflict network) over time based on proposed model. Since the coefficient of geopolitical relationships network on cobalt upstream trade network is not significant in most years, only the results of middle and downstream are shown in Fig. 12.

Fig. 12 shows that the absolute value of coefficient in downstream is larger than that of midstream, which indicates that the influence of geopolitical relationship on cobalt downstream trade is greater than that on the midstream. The downstream market of cobalt is larger and has more participants than the midstream market (van den Brink et al., 2020), so it is more vulnerable to geopolitical relationships. What's more, the absolute value of coefficient for cooperation network is larger than conflict network, which denotes that the cooperation between the two countries has a greater impact on the establishment of trade relations. A possible explanation for this phenomenon is the geopolitical cooperation can provide mechanisms and platforms for resolving trade disputes through dialogue, negotiation, and arbitration, reducing the negative impact of trade frictions (Hartono, 2022). In other words, geopolitical cooperation can reduce the negative effects of geopolitical conflicts to a certain extent. In addition, the coefficient value of

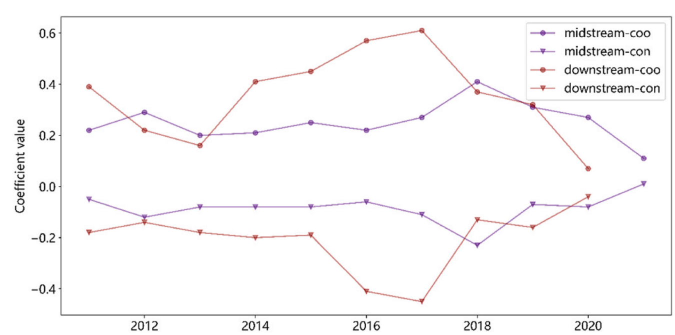

Fig. 12. The evolution of the impact coefficient of the geopolitical relationship network.
The effect of geopolitical conflict on cobalt trade is significantly negative, so the curve is lower on the x-axis.

Table 4
The estimation results of the ERGM for basic and proposed model.

	Basic Model			Proposed Model		
	Upstream	Midstream	Downstream	Upstream	Midstream	Downstream
edges	−7.91*** (0.271)	−3.83*** (0.039)	−2.14*** (0.021)	−6.84*** (0.346)	−3.09*** (0.053)	−1.57*** (0.031)
Nodecov(GDP)	0.47*** (0.091)	0.6*** (0.015)	0.7*** (0.012)	0.43*** (0.09)	0.6*** (0.015)	0.7*** (0.012)
Nodecov(FDI)	0.17** (0.055)	0.08*** (0.012)	0.14*** (0.012)	0.15* (0.063)	0.08*** (0.013)	0.16*** (0.012)
Nodecov(POP)	0.25*** (0.05)	0.34*** (0.011)	0.39*** (0.015)	0.26*** (0.053)	0.36*** (0.011)	0.39*** (0.014)
Edgecov(DIST)				−1.8*** (0.416)	−1.21*** (0.055)	−0.76*** (0.029)
Edgecov(COL)				0.24 (0.127)	0.02 (0.024)	0 (0.015)
Edgecov(CGB)				0.05 (0.067)	0.21*** (0.014)	0.13*** (0.013)
Edgecov(COO)				0.08 (0.049)	0.27*** (0.033)	0.32*** (0.045)
Edgecov(CON)				−0.03 (0.065)	−0.08** (0.025)	−0.16*** (0.028)
AIC	621.8786627	15191.66417	34716.53934	593.7869345	13977.3105	33572.30101
BIC	655.7914186	15225.57693	34750.45209	670.0906353	14053.6142	33648.60471

Notes: *, **, and *** indicate significance levels of 10%, 5%, 1%, respectively. The value in parentheses are standard deviations. Same as the following table.

geopolitical relationship network increases first and then decreases after 2017, which means that the influence of geopolitical relationships on the formation of cobalt trade has become smaller and smaller after 2017. Before 2017, the DRC has been plagued by political instability, violence, and human rights abuses, which have created uncertainty and risks for cobalt mining and trade (Sovacool, 2019). Geopolitical conflicts and tensions over the control and exploitation of cobalt resources in the DRC could affect the global cobalt trade and lead to supply disruptions, price volatility, and market uncertainties. In addition, there might have been strategic alliances and partnerships formed between cobalt-producing countries and cobalt-consuming countries to facilitate trade and ensure access to resources before 2017. These agreements could have influenced the cobalt trade network and increased the importance of geopolitical relations. However, after 2017, some countries have started to diversify their cobalt supply sources to reduce their dependence on the DRC (Earl et al., 2022). For example, China has been investing in cobalt mining and exploration projects in other countries such as the Philippines, Indonesia, and Canada. This diversification could reduce the geopolitical risks and tensions associated with the cobalt trade and create a more stable and balanced cobalt market.

In order to analyze the influence of the status of countries in the geopolitical relationship network on the evolution of cobalt trade network, we add degree centrality and weighted degree centrality of geopolitical relationship network into ERGM model as model 1 and model 2. Table 5 shows the results of model 1 and model 2.

From the result of model 1, the degree centrality of the cooperation and conflict network have significant impact on the evolution of cobalt trade relations only in the midstream and downstream, indicating that geopolitical cooperation with more countries is conducive to the formation of cobalt trade relations in midstream and downstream. Countries usually cooperate with others to ensure the safety of cobalt supply, and they will not sever the geopolitical cooperation that have been established before (Li et al., 2022b). In terms of model 2, the weighted centrality of the cooperation network has a significant positive impact on the establishment of trade relations in the whole industrial chain, while the significance of the upstream is obviously weaker than that of midstream and downstream. It means that the total intensity of cooperation and conflict can affect the evolution of cobalt trade network, and upstream of cobalt trade are insensitive to geopolitical relations, which consistent with the results of proposed model.

6. Conclusion and policy implications

This paper explores the evolution of geopolitical relationships network and cobalt trade network, and discusses the impact of geopolitical relationships network on the evolution of cobalt trade network from the industrial chain perspective. Based on global geopolitical relationships and trade data, this research provides the first evidence on the impact of geopolitical relationships on cobalt trade network, critically discussed its heterogeneity impact on different stages of industrial chain. The main conclusions are as followed.

First, the total size of cooperation network is larger than that of conflict network, and more and more countries have participated in the global geopolitical conflict. The density of cooperation network is higher than that of conflict network, and the average shortest path of network is gradually decreasing, the clustering coefficient is gradually increasing, and the average degree and reciprocity are gradually increasing. The United States plays the most important role in the cooperative network and the conflict network, as it has the largest number of conflict relations and cooperation relations as the same time. Considering the change in geopolitical relationships network pattern and the special status of the United States, countries should handle international relations with caution to cope with increasing number of international conflicts. Meanwhile, countries should maintain reasonable relationships with the United States to ensure their own interests to the greatest extent.

Second, the trade pattern of cobalt in the midstream and downstream has fluctuated violently in recent years. The average degree and average weighted degree of midstream and downstream network dropped sharply in 2020, while that of upstream kept relatively stable. Besides, downstream trade is the most efficient and stable, and the trade efficiency of upstream is getting better. Congo dominates cobalt upstream trade with its excellent resource endowment. China and Congo are important countries in midstream trade. The United States, Germany, and many European countries keep the significant role in downstream trade. Therefore, countries should strengthen contacts with the midstream and downstream of cobalt exporting countries and demand countries, to cope with the drastic changes caused by the epidemic in recent years.

Third, geopolitical relationships only affect the evolution of midstream and downstream trade network for cobalt, and that of upstream trade network is insensible to geopolitical relationships. The cooperation network has significant positive impact only on the evolution of midstream and downstream, and the conflict network has the opposite impact on them. In addition, the degree centrality of the cooperation and conflict network have significant impact on the evolution of cobalt trade relations only in the midstream and downstream. Indicating that if countries want to establish cobalt midstream and downstream trade with more countries, they must establish geopolitical cooperation with more countries and reduce geopolitical conflicts at the same time.

Finally, the influence of geopolitical cooperation on the evolution of cobalt trade network is greater than that of geopolitical conflict. In addition, the downstream trade network is more susceptible to

Table 5
The results of ERGM for model 3 and model 4.

	Model 1			Model 2		
	Upstream	Midstream	Downstream	Upstream	Midstream	Downstream
edges	−8.52*** (0.586)	−5.15*** (0.092)	−3.43*** (0.051)	−7.21*** (0.378)	−3.32*** (0.056)	−1.72*** (0.032)
Nodecov(GDP)	0.2 (0.119)	0.48*** (0.017)	0.53*** (0.012)	0.34** (0.1)	0.58*** (0.016)	0.67*** (0.012)
Nodecov(FDI)	−0.23** (0.073)	−0.13*** (0.016)	0.04** (0.013)	−0.25** (0.091)	−0.08*** (0.018)	0.1*** (0.013)
Nodecov(POP)	0.22** (0.065)	0.25*** (0.013)	0.22*** (0.013)	0.31*** (0.059)	0.33*** (0.012)	0.34*** (0.014)
Nodecov(dcoo)	1.09 (0.76)	3*** (0.124)	3.79*** (0.088)			
Nodecov(dcon)	3.69* (1.423)	−1.25*** (0.25)	−3.84*** (0.19)			
Nodecov(wdcoo)				4.9* (2.089)	6.58*** (0.404)	7.71*** (0.349)
Nodecov(wdcon)				−0.29 (2.168)	−4.63*** (0.429)	−7.18*** (0.395)
Edgecov(DIST)	−1.66*** (0.442)	−1.02*** (0.058)	−0.57*** (0.031)	−1.72*** (0.428)	−1.18*** (0.056)	−0.74*** (0.029)
Edgecov(COL)	0.3* (0.135)	0.15*** (0.025)	0.14*** (0.016)	0.26 (0.134)	0.07** (0.024)	0.03* (0.015)
Edgecov(CGB)	0.03 (0.068)	0.22*** (0.015)	0.14*** (0.014)	0.05 (0.068)	0.21*** (0.014)	0.13*** (0.013)
Edgecov(COO)	0 (0.051)	0 (0.019)	−0.03 (0.021)	0.05 (0.048)	0.1** (0.029)	0.11** (0.039)
Edgecov(CON)	−0.04 (0.053)	−0.01 (0.018)	−0.03 (0.019)	−0.05 (0.056)	−0.02 (0.023)	−0.06* (0.026)
AIC	540.1275004	12470.22013	30258.57278	574.9172035	13659.9958	33034.62034
BIC	633.3875791	12563.48021	30351.83286	668.1772822	13753.25588	33127.88042

geopolitical relationships. Further, the influence of geopolitical relationships on the formation of cobalt trade has become smaller and smaller after 2017. Therefore, countries should pay special attention to the geopolitical cooperation and establish cooperation with as many countries as possible to promote cobalt trade. Meanwhile, all countries should make concerted efforts to deal with the epidemic, so as to enhance the positive effect of cooperation on trade.

CRediT author statement

Yaoqi Guo: Conceptualization, Methodology, Supervision Yingli Li: Investigation, Formal analysis, Data curation Yongheng Liu: Software, Validation, Writing - Original Draft Hongwei Zhang: Resources, Writing - Reviewing and Editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We gratefully acknowledge financial support from the Post-Funded Projects of the National Social Science Foundation (No. 22FGLB032), National Natural Science Foundation of China (Nos. 72204273 and 42071260), the Humanities and Social Science Research Project of Ministry of Education of China (No. 22YJCZH235), and the National Social Science Foundation of China (Nos. 22&ZD098 and 22CTJ021).

Appendix

Table A1
Commodity Customs Code

Hs code	Commodity	Category	Hs code	Commodity	Category
260500	Cobalt ores and concentrates	Ore	845951	Boring machines nes for removing metal	Product
282200	Cobalt oxides and hydroxides; commercial cobalt oxides	Material	845959	Milling mach, knee type nes for removing metal	Product
282739	Cobalt chloride	Material	845961	Milling machines nes, numerically controlled for removing metal	Product
283329	Sulphates (excl. of 2833.21–2833.27)	Material	845969	Milling machines nes, for removing metal	Product
283699	Carbonates; n.e.c. in heading no. 2836 and other than lithium or strontium	Material	845970	Threading or tapping machines nes for removing metal	Product
291529	Cobalt oxalate	Material	846021	Num controlled grinding machines nes, accuracy 0.01 mm	Product
7501	Nickel matte, interim products of nickel metallurgy	Material	846029	Grinding machines nes, accurate to 0.01 mm	Product
810520	Cobalt, unwrought, matte & other intermediate products, powders	Material	846031	Sharpening (tool or cutter grinding) mach n/c for removing metal	Product
810590	Cobalt; articles n.e.c. in heading no. 8105	Material	846039	Sharpening (tool or cutter grinding) mach nes for removing metal	Product
8802	Aircraft; helicopters, airplanes, spacecraft	Product	846040	Honing or lapping machines for removing metal	Product
284990	Carbides; whether or not chemically defined, other than of calcium or silicon	Product	846090	Mach-tools for deburring polishing etc for fin met nes o/t hdg 84.61	Product
320417	Dyes; pigments and preparations based thereon	Product	846120	Shaping or slotting machines by removing metal	Product
320710	Pigment; Pigments, sunscreens, pigments and similar preparations prepared	Product	846130	Broaching machines by removing metal	Product
320720	Enamel and glazes	Product	846140	Gear cutting, gear grindg or gear finishg machines by removg metal	Product
320740	Glass; Powder glass and other glass in the form of powder, granule, or sheet	Product	846150	Sawing or cutting-off machines by removing metal	Product
381511	Supported catalysts, with nickel/nickel compounds as the active subst	Product	847190	A magnetic or optical reader, a machine for transcribing data in coded form onto a data medium and for processing such data	Product
381512	Supported catalysts precious metal/compds thereof as the active subs	Product	850110	A motor whose output power does not exceed 37.5W	Product
381519	Catalysts, supported; reaction initiators, reaction accelerators and catalytic preparations, with an active substance other than nickel or precious metals or their compounds, n.e.c. or included	Product	850511	Magnets; permanent magnets and articles intended to become permanent magnets after magnetisation, of metal	Product
381590	Reaction initiators, reaction accelerator&catalytic preparations, nes	Product	850650	Lithium primary cells and batteries	Product
711510	Catalysts in the form of wire cloth or grill, of platinum	Product	850730	Electric accumulators; nickel-cadmium, including separators, whether or not rectangular (including square)	Product
720299	Ferro-alloy; n.e.s in heating	Product	850780	Electric accumulators, nes	Product
840710	Engines; for aircraft, spark-ignition reciprocating or rotary internal combustion piston engines	Product	851010	Shaver with electric motor	Product
840910	Engines; parts of aircraft engine	Product	851020	A haircut; With a separate motor	Product
841111	Turbo-jets; of a thrust not exceeding 25 kN	Product	851030	Depilation appliances; Independent motor	Product
841112	Turbo-jets; of a thrust exceeding 25 kN	Product	851390	Electric lamps, designed to function by their own source of energy	Product
841121	Turbo propeller; of a power not exceeding 1100kw	Product	851711	cordless handsets	Product
841122	Turbo propeller; of a power exceeding 1100kw	Product	851712	Telephones for cellular networks or for other wireless networks	Product
841181	Turbines; gas turbines, of a power not exceeding 5000 KW	Product	851821	The speaker; Single, mounted in its housing	Product
841182	Turbines; gas turbines, of a power exceeding 5000 KW	Product	851822	The speaker; Multiple, installed in the same case	Product
841191	Turbines; parts of turbo-jets and turbo propellers	Product	851829	The speaker; Not installed in its housing	Product
841199	Turbines; excluding turbo-jets and turbo propellers	Product	852530	Television cameras	Product
845710	Machining centers, for working metal	Product	870321	Automobiles, spark ignition engine of <1000 cc	Product
845720	Unit construction machines (single sta) for working metal	Product	870322	Automobiles w reciprocate piston engine displacg >1000 cc–1500 cc	Product
845730	Multi-station transfer machines for working metal	Product	870323	Automobiles w reciprocate piston engine displacg >1500 cc–3000 cc	Product

(continued on next page)

Table A1 (*continued*)

Hs code	Commodity	Category	Hs code	Commodity	Category
845811	Horizontal lathes numerically controlled for removing metal	Product	870324	Automobiles with reciprocating piston engine displacing >3000 cc	Product
845819	Horizontal lathes nes for removing metal	Product	870331	Automobiles with diesel engine displacing not more than 1500 cc	Product
845891	Lathes nes numerically controlled for removing metal	Product	870332	Automobiles with diesel engine displacing more than 1500 cc–2500 cc	Product
845899	Lathes nes for removing metal	Product	870333	Automobiles with diesel engine displacing more than 2500 cc	Product
845910	Way-type unit head mches for removing metal	Product	8802	Aircraft; helicopters, airplanes, spacecraft	Product
845921	Drilling mches nes; numerically controlled for removing metal	Product	902121	Dental accessories; Denture Products - Others	Product
845929	Drilling mches nes, for removing metal	Product	902131	An artificial part of the human body	Product
845931	Boring-milling mches nes, numerically controlled for removing metal	Product	902139	An artificial part of the human body; It does not include artificial joints	Product
845939	Boring-milling mches nes for removing metal	Product			

Table A2
Names and ISO codes of countries.

Country	ISO code	Country	ISO code	Country	ISO code	Country	ISO code	Country	ISO code
Algeria	DZA	Cote d'Ivoire	CIV	Iceland	ISL	Malaysia	MYS	South Korea	KOR
Algeria	DZA	Ecuador	ECU	India	IND	Moldova	MDA	Spain	ESP
Angola	AGO	Egypt	EGY	Indonesia	IDN	Mongolia	MNG	Sri Lanka	LKA
Argentina	ARG	Ethiopia	ETH	Iran	IRN	Mozambique	MOZ	Sudan	SDN
Aruba	ABW	Fiji	FJI	Iraq	IRQ	Myanmar	MMR	Suriname	SUR
Australia	AUS	Finland	FIN	Jamaica	JAM	Netherlands(Holland)	NLD	Syria	SYR
Austria	AUT	France	FRA	Japan	JPN	Nicaragua	NIC	Tanzania	TZA
Belgium	BEL	Gambia	GMB	Jordan	JOR	Niger	NER	Uganda	UGA
Bolivia	BOL	Georgia	GEO	Kazakhstan	KAZ	Nigeria	NGA	Ukraine	UKR
Brazil	BRA	Germany	DEU	Korea, DPR	PRK	Oman	OMN	United Kingdom	GBR
Cameroon	CMR	Ghana	GHA	Kuwait	KWT	Pakistan	PAK	United States	USA
Canada	CAN	Greece	GRC	Latvia	LVA	Panama	PAN	Venezuela	VEN
Chile	CHL	Guatemala	GTM	Lebanon	LBN	Papua New Guinea	PNG	Yemen	YEM
China	CHN	Guinea	GIN	Liberia	LBR	Romania	ROM	Zambia	ZMB
Colombia	COL	Guyana	GUY	Libya	LBY	Sierra Leone	SLE	Zimbabwe	ZWE
Comoros	COM	Haiti	HTI	Lithuania	LTU	Somalia	SOM		
Congo-kinshasa	COD	Hong Kong	HKG	Madagascar	MDG	South Africa	ZAF		

References

Abbassi, W., Kumari, V., Pandey, D.K., 2022. What makes firms vulnerable to the Russia–Ukraine crisis? J. Risk Finance (ahead-of-print).

Alam, M.K., Tabash, M.I., Billah, M., Kumar, S., Anagreh, S., 2022. The impacts of the Russia–Ukraine invasion on global markets and commodities: a dynamic connectedness among G7 and bric markets. J. Risk Financ. Manag. 15, 352.

Allen, T., Arkolakis, C., 2014. Trade and the topography of the spatial economy. Q. J. Econ. 129, 1085–1140.

Asiedu, E., 2006. Foreign direct investment in Africa: the role of natural resources, market size, government policy, institutions and political instability. World Econ. 29, 63–77.

Barrat, A., Barthélemy, M., Pastor-Satorras, R., Vespignani, A., 2004. The architecture of complex weighted networks. Proc. Natl. Acad. Sci. USA 101, 3747–3752.

Bernauer, T., Böhmelt, T., 2020. International conflict and cooperation over freshwater resources. Nat. Sustain. 3, 350–356.

Catoto Capitango, J.A., Garat de Marin, M.S., Soriano Flores, E., Rojo Gutiérrez, M.A., Gracia Villar, M., Durántez Prados, F.Á., 2022. Inequalities and asymmetries in the development of Angola's provinces: the impact of colonialism and civil war. Soc. Sci. 11, 334.

de Soyres, F., Mulabdic, A., Murray, S., Rocha, N., Ruta, M., 2019. How much will the Belt and Road Initiative reduce trade costs? International Economics 159, 151–164.

Earl, C., Shah, I.H., Cook, S., Cheeseman, C.R., 2022. Environmental sustainability and supply resilience of cobalt. Sustainability 14, 4124.

Federo, R., Bustamante, X., 2022. The ties that bind global governance: using media-reported events to disentangle the global interorganizational network in a global pandemic. Soc. Network. 70, 253–266.

Fink, C., Mattoo, A., Neagu, I.C., 2005. Assessing the impact of communication costs on international trade. J. Int. Econ. 67, 428–445.

Gong, X., Fu, C., Huang, Q., Lin, M., 2022a. International political uncertainty and climate risk in the stock market. J. Int. Financ. Mark. Inst. Money 81, 101683.

Gong, X., Jin, Y., Sun, C., 2022b. Time-varying pure contagion effect between energy and nonenergy commodity markets. J. Futures Mark. 42, 1960–1986.

Gong, X., Lin, B., 2021. Effects of structural changes on the prediction of downside volatility in futures markets. J. Futures Mark. 41, 1124–1153.

Gong, X., Sun, Y., Du, Z., 2022c. Geopolitical risk and China's oil security. Energy Pol. 163, 112856.

Gupta, R., Gozgor, G., Kaya, H., Demir, E., 2019. Effects of geopolitical risks on trade flows: evidence from the gravity model. Eurasian Economic Review 9, 515–530.

Gustafsson, R., Dutta, A., Bouri, E., 2022. Are energy metals hedges or safe havens for clean energy stock returns? Energy 244, 122708.

Habib, K., Hamelin, L., Wenzel, H., 2016. A dynamic perspective of the geopolitical supply risk of metals. J. Clean. Prod. 133, 850–858.

Hartono, A.C., 2022. The impact of geopolitical influence on regional cooperation and integration in East Asia. Asian Journal of Comparative Politics 7, 353–375.

Hoekman, B., Nicita, A., 2011. Trade policy, trade costs, and developing country trade. World Dev. 39, 2069–2079.

Huang, J., Ding, Q., Wang, Y., Hong, H., Zhang, H., 2021. The evolution and influencing factors of international tungsten competition from the industrial chain perspective. Resour. Pol. 73, 102185.

Jagtap, S., Trollman, H., Trollman, F., Garcia-Garcia, G., Parra-López, C., Duong, L., Martindale, W., Munekata, P.E., Lorenzo, J.M., Hdaifeh, A., 2022. The Russia-Ukraine conflict: its implications for the global food supply chains. Foods 11, 2098.

Jiang, H., Yu, M., 2021. Understanding RCEP and CPTPP: from the perspective China's dual circulation economic strategy. China Econ. J. 14, 144–161.

Lee, L.W., Sharma, P., Barnes, B.R., 2022. Adopting big data to create an "outside-in" global perspective of guanxi. J. Bus. Res. 139, 614–628.

Levin, N., Ali, S., Crandall, D., 2018. Utilizing remote sensing and big data to quantify conflict intensity: the Arab Spring as a case study. Appl. Geogr. 94, 1–17.

Li, M., Yuan, W., Yuan, W., Niu, F., Li, H., Hu, D., 2022a. Spatio-temporal evolution and influencing factors of geopolitical relations among Arctic countries based on news big data. J. Geogr. Sci. 32, 2036–2052.

Li, Y., Huang, J., Zhang, H., 2022b. The impact of country risks on cobalt trade patterns from the perspective of the industrial chain. Resour. Pol. 77, 102641.

Lohner, J., Neverla, I., Banjac, S., 2019. Conflict-sensitive journalism? Journalistic role perceptions and practices in democratization conflicts. In: Voltmer, K., Christensen, C., Neverla, I., Stremlau, N., Thomass, B., Vladisavljević, N., Wasserman, H. (Eds.), Media, Communication and the Struggle for Democratic Change: Case Studies on Contested Transitions. Springer International Publishing, Cham, pp. 59–81.

Martínez-Zarzoso, I., Chelala, S., 2021. Trade agreements and international technology transfer. Rev. World Econ. 157, 631–665.
Meng, F., Zhao, Y., 2022. How does digital economy affect green total factor productivity at the industry level in China: from a perspective of global value chain. Environ. Sci. Pollut. Control Ser. 29, 79497–79515.
Monge, M., Gil-Alana, L.A., de Gracia, F.P., 2017. Crude oil price behaviour before and after military conflicts and geopolitical events. Energy 120, 79–91.
Nejjari, N., Lahlou, S., Fadi, O., Zkik, K., Oudani, M., Benbrahim, H., 2021. Conflict Spectrum: an Empirical Study of Geopolitical Cyber Threats from a Social Network Perspective, 2021 Eighth International Conference on Social Network Analysis, Management and Security (SNAMS). IEEE, pp. 1–7.
Nerlinger, M., Utz, S., 2022. The Impact of the Russia-Ukraine Conflict on the Green Energy Transition–A Capital Market Perspective (Swiss Finance Institute Research Paper).
Ngoc, N.M., Viet, D.T., Tien, N.H., Hiep, P.M., Anh, N., Anh, L., Truong, N., Anh, N., Trung, L., Dung, V., 2022. Russia-Ukraine war and risks to global supply chains. Int. J. Mech. Eng. 7, 633–640.
Pollins, B.M., 1989. Conflict, cooperation, and commerce: the effect of international political interactions on bilateral trade flows. Am. J. Polit. Sci. 737–761.
Saz-Carranza, A., Vandendriessche, M., Nguyen, J., Agell, N., 2022. The EU's interactions with formal intergovernmental organizations: a big data analysis of news media. J. Eur. Integrat. 44, 635–655.
Sheng, L., do Nascimento, D.F., 2021. The Belt and Road Initiative in South–South Cooperation: the Impact on World Trade and Geopolitics. Springer Nature.
Shi, Q., Sun, X., Xu, M., Wang, M., 2022. The multiplex network structure of global cobalt industry chain. Resour. Pol. 76, 102555.
Solingen, E., 2021. Geopolitics, Supply Chains, and International Relations in East Asia. Cambridge University Press.
Sovacool, B.K., 2019. The precarious political economy of cobalt: balancing prosperity, poverty, and brutality in artisanal and industrial mining in the Democratic Republic of the Congo. Extr. Ind. Soc. 6, 915–939.
Sun, X., Hao, H., Liu, Z., Zhao, F., Song, J., 2019. Tracing global cobalt flow: 1995–2015. Resour. Conserv. Recycl. 149, 45–55.
van den Brink, S., Kleijn, R., Sprecher, B., Tukker, A., 2020. Identifying supply risks by mapping the cobalt supply chain. Resour. Conserv. Recycl. 156, 104743.
Vivoda, V., 2011. Determinants of foreign direct investment in the mining sector in Asia: a comparison between China and India. Resour. Pol. 36, 49–59.
Wei, L., Xiao, J., Huang, Y., 2020. Is there a segmentation inside global value chain of the automobile industry in China? Am. J. Ind. Bus. Manag. 10, 374.
Wen, W., Yang, S., Zhou, P., Gao, S.Z., 2021. Impacts of COVID-19 on the electric vehicle industry: evidence from China. Renew. Sustain. Energy Rev. 144, 111024.
Xu, J., Yang, X., Razzaq, A., 2021. Understanding the role of humanistic factors in trade network evolution across the Belt and Road initiative countries using the exponential random graph model. Complexity 2021, 1961061.
Zheng, S., Zhou, X., Tan, Z., Zhang, H., Liu, C., Hao, H., Hu, H., Cai, X., Yang, H., Luo, W., 2022. Preliminary study on the global impact of sanctions on fossil energy trade: based on complex network theory. Energy for Sustainable Development 71, 517–531.
Zuo, Z., McLellan, B.C., Li, Y., Guo, H., Cheng, J., 2022. Evolution and insights into the network and pattern of the rare earths trade from an industry chain perspective. Resour. Pol. 78, 102912.